

EXPERIMENT 3

Stock Price Prediction using Linear Regression

1. Problem Definition

The objective of this experiment is to predict the stock's closing price using the previous day's closing price.

Since closing price is continuous, this is a Regression Problem and also a Time-Series Regression task.

2. Data Understanding

The dataset contains daily stock price data including:

- Date
- Open
- High
- Low
- Close
- Volume

Each row represents one trading day.

Target Variable (y)

- Close (Current Day)

3. Feature Engineering

A new column was created:

- Previous_Price

This represents:

$\text{Previous_Price} = \text{Closet}_{-1}$

The first row (with no previous value) was removed.

4. Train-Test Split

Since this is time-series data, splitting was done sequentially:

- First 80% → Training
- Last 20% → Testing

Random shuffling was not performed to preserve chronological order.

Regression Model

Linear Regression was used to model:

$\text{Closet} = b_0 + b_1(\text{Previous_Page})$

Model Training

The model learned coefficients by minimizing Mean Squared Error on training data.

5. Prediction and Evaluation

The trained model predicted stock closing prices for the test dataset.

Mean Squared Error (MSE)

$\text{MSE} = 6.756204507398451$

This indicates the average squared prediction error.

R² Score

$R^2 = 0.9710074537224115$

This means:

- The model explains approximately 97.10% of the variance in stock prices.
- There is a strong linear relationship between previous and current prices.